

John Sipahioglu

Instructor - Computer Science - Stark State College

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Department of Computer Science and Information Systems

Stark State College

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Education

- Ph.D. - Computer Science - Kent State University - In-progress
- M.S. - Computer Science - Kent State University - August 2024
- B.S. - Computer Science - Kent State University - May 2024
- B.A. - Mathematics - Kent State University - May 2024

Advisors

Prof. Jonathan I. Maletic (Ph.D., 2024 - Present)

Dr. Younghun Chae (M.S., 2023 - 2024)

Research Interests

Code reuse, design recovery, code quality, software evolution, technical debt, and maintenance

I am currently contributing to the srcML infrastructure, which supports source code fact extraction, querying, transformation, and syntactic differencing. I am also exploring ways to use the srcML infrastructure to identify technical debt in systems.

Projects

- srcML - Contribute to writing grammars to add new languages to srcML and presently implementing parsing for Rust. The srcML tool wraps source code elements in XML tags to embed the abstract syntax tree into the source code. srcML serves as the base for other tools to query, difference, parse, and transform source code.
- srcBind - A tool for creating idiomatic bindings and wrappers of C (and eventually Rust) libraries in object-oriented languages, such as Python, C++ and Java.
- srcDispatch - Contribute to continued development and support of more language features in srcDispatch. srcDispatch is a SAX event dispatcher for srcML.

Academic Experience

- *Instructor, Full Time: Computer Science* - Stark State College (August 2025 - Present)
- *Part Time Faculty: Computer Science* - Kent State University at Stark (August 2024 - Present)
- *Adjunct Instructor: Computer Science* - Stark State College (August 2024 - August 2025)
- *Tutor: Computer Science, Mathematics, Physics, Chemistry* - Kent State University at Stark (January 2022 - May 2024)

Industry Experience

- *IT Coop* - PCC Airfoils (July 2022 - January 2023, June 2023 - August 2024)

Funding Awards

- Summer Research Worker (May 2025 - August 2025) - Funded under NSF grant CNS 20-16465. Created markup specification to add the Rust programming language to srcML and contributed to srcDispatch to parse srcML
- Ohio Space Consortium Grant (August 2023 - May 2024) - Awarded \$10,000 for research into potential benefits of updating protocols for securing Deep Space Network
- Choose Ohio First - Computer Science Scholarship (January 2022 - May 2024) - Awarded funds for undergraduate tuition as part of the COF-CS program

Publications and Scholarly Works

Works Under Review or Near Submission

- Al-Ramadan, F., Sipahioglu, J.A., Behler, J.A.C., Dragan, N., Decker, M.J., Collard, M.L., Maletic, J.I., “Should this Method be Static? Automatic Identification of Static Method Candidates”, submitted May 16th, 2026 to IEEE Transactions on Software Engineering (TSE).
- Sipahioglu, J.A., Al-Ramadan, F., Rossi, K.N., Behler, J.A.C., Decker, M.J., Collard, M.L., Maletic, J.I., “Reverse Engineering Object-Oriented Designs from C Systems”, to be submitted Fall 2026 to the 34th IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER).

Accepted and to Appear

- Sipahioglu, J.A., Behler, J.A.C., Testa, S., Al-Ramadan, A., Decker, M.J., Collard, M.L., Maletic, J.I., (2026) “nameCollector: Extracting Source Code Identifiers Along with their Types and Context”, in the Proceedings of 42nd IEEE International Conference on Software Maintenance & Evolution (ICSME), Tool Demonstrations Track, Benevento, Italy, Sept 14 - 18, 5 pages. (TO APPEAR)

Publications (Refereed)

N/A

Theses

- Sipahioglu, J.A., “Modernizing Deep Space Network Security”, M.S. Thesis, Kent State University, Department of Computer Science, August 2024.

Professional Activities/Services

Conference Reviewer

- 2026 14th IEEE Working Conference on Software Visualization (VISOFT) - Additional Reviewer
- 2026 42nd IEEE International Conference on Software Maintenance and Evolution (ICSME '26) - Additional Reviewer
- 2026 34th IEEE/ACM International Conference on Program Comprehension (ICPC) - Additional Reviewer

University Activities/Services

- 2025 Choose Ohio First (COF) Poster Competition at Kent State University - Judged 6 computer science posters
- 13th International Conference on Intelligent Human Computer Interaction (IHCI 2021) at Kent State University - Assisted with High School outreach workshop

Teaching

Stark State College

Summer 2026

- CSE233 - C++ Programming - CSE234
- Advanced C++ Programming - CSE235
- Python Development

Spring 2026

- CSE121 - Mobile Development Architecture
- CSE122 - Programming Logic and Problem Solving
- CSE233 - C++ Programming
- CSE234 - Advanced C++ Programming
- CFS137 - Computer Crime and Investigation
- CFS258 - Cyber Forensics and Data Recovery
- CFS287 - Network Forensics

Fall 2025

- CSE122 - Programming Logic and Problem Solving
- CSE233 - C++ Programming
- NET220 - UNIX/Linux Operating Environments
- CFS210 - Cyber Defense Fundamentals
- CFS257 - File System Analysis
- CFS258 - Cyber Forensics and Data Recovery

Summer 2025

- WDD226 - Web Development with PHP and MySQL

Spring 2025

- CFS257 - File System Analysis

- CFS258 - Cyber Forensics and Data Recovery

- CFS287 - Network Forensics

Fall 2024

- CFS137 - Computer Crime and Investigation

Kent State University at Stark

Spring 2025

- CS13001 Computer Science I - Programming and Problem Solving (Lab Section)

- CS23001 Computer Science II - Data Structures and Abstraction (Lab Section)

Fall 2024

- CS13001 Computer Science I - Programming and Problem Solving (Lab Section)

- CS23001 Computer Science II - Data Structures and Abstraction (Lab Section)